Evaluating Quantum Contextuality as an Error Mitigation Resource in the Mermin–Peres Magic Square Game

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Introduction

Background

Quantum computers offer solutions to problems that classical computers cannot solve. The current generations of quantum computers (Noisy Intermediate-Scale Quantum/NISQ) are unreliable due to noise caused by limited qubit counts, gate errors, and measurement issues that degrade any algorithm run on these devices. There are two main ways to reduce the error: increase qubits counts or build noise-resistant architecture. Since it is incredibly hard to scale qubits, mitigating these issues through noise-aware algorithms remains the most practical option.

Contextuality is a fundamental property of quantum mechanics, describing how measurement outputs can be dependent on the compatible measurements done alongside it. In classical systems, objects can be assigned a value prior to observation since we assume objects to exist independently of its measurement context. In contrast, a quantum system's qubits don't possess a definite value before measurement; rather, its value is fundamentally dependent on the mathematical context of its measurement. Researchers have identified contextuality as a key resource in improving error rates in future NISQ devices as the first step towards universal quantum calculation.

Purpose of Study and Research Question

The purpose of this study is to determine how contextuality can influence how the Mermin-Peres magic square algorithm functions with noisy quantum computers. The magic square is a well-known illustration of contextuality because even though the magic square provides many observable outcomes from the operators measured, no classical, non-contextual, theory can predict these outcomes. This study aims to answer the question: "To what extent does

preserving contextuality improve the accuracy of the Mermin–Peres magic square algorithm on noisy quantum hardware?"

Key Definitions and Terminology

The term "contextuality" refers to the inability to define "measurement values" based on an a priori assignment of those values nor the existence of the context from which they arise. "Magic" resources refer to non-stabilizer quantum resources used in computations that are not standard binary. The measure of how close the actual result of the computation is to what should have theoretically happened is known as the "fidelity" of the calculation. The calculation of the fidelity is almost like that of the "trace distance," which provides a numerical measure of how similar two probability distributions are.

NISQ devices are quantum computers that do not possess the complete set of resources needed to implement error-correction schemes, nor do they have resistance to environmental noise or decay; therefore, they are referred to as "Noisy Intermediate-Scale Quantum" (NISQ) devices.

In quantum computing, "error mitigation" is a technique for reducing the effects of noise on quantum circuits. One technique for error mitigation is zero-noise extrapolation (ZNE), which allows for the correlation of an experimental result to a theoretical result (if there is one) from measurements on computations run at several different levels of noise.

Contextuality as a Computational Resource

According to Howard et al.'s (2014) interpretation, it is contextuality that gives rise to the "magic" that is required for universal quantum computation. Furthermore, contextuality contributes non-classical correlation that makes computational speed up possible, meaning that although there are no classical correlations to provide quantum speed-up, entanglement will not

exist within the same way that contextuality presents an advantage for quantum computation (Mermin & Peres, 1990).

Howard et al. provide a direct operational test of contextuality through their use of the Mermin–Peres magic square. Furthermore, Howard et al. continue to explain that maintaining the contextuality of quantum circuits may be necessary in order for the circuits to provide a quantum advantage. Within NISQ devices, maintaining the contextuality of any quantum circuit can be essential because noise can degrade any non-classical resources before computations can be performed.

Error Mitigation on Noisy Quantum Devices

In their paper published in 2019 by Kandala, et al., they described the limitations of (NISQ) analog quantum computing systems, including how they are limited in their use as simulators of quantum systems (the field of Quantum Chemistry) through the use of the Variational Quantum Eigensolver (VQE), which is a type of quantum circuit that uses classical and quantum feedback to estimate the ground state energy (the lowest energy state) of a given molecule. Due to high amounts of noise from gate error, decoherence, and measurement uncertainty, the accuracy of their simulations on NISQ hardware was significantly limited. Thus, Kandala et al. proposed methods for reducing errors using error mitigation techniques, including ZNE, whereby the same quantum circuit is run multiple times at different noise levels and extrapolated to a zero-noise result through methods such as gate folding or Richardson extrapolation. They demonstrated how these techniques improve the accuracy of observables measured on NISQ devices compared to idealized results.

However, the methodology used to enhance numeric accuracy did not include an evaluation of whether the non-classical computational resources used to produce the data were

preserved. Contextuality and Magic were not evaluated by Kandala et al. after performing measurement correction and reconstruction of expectation values, leading to an uncertain determination as to whether improvements in expectations represent actual preservation of contextual correlations or simply noise reduction at the output level. This distinction is essential because no reliability can be placed on any increase in performance due to no evidence that a quantum advantage has continued.

Magic and Resource Degradation Under Noise

The authors of a new paper, Haug and colleagues (2023), have extended the resource-theory framework to create the first scalable measures of magic (i.e., a quantifiable resource associated with non-stabilizer states and contextuality). These measures allow quantification of the amount of magic in any quantum state, as well as how noise affects magic. Some noise channels diminish magic significantly more than others, thus decreasing the quantum computing advantage that comes from having magic.

While Haug and colleagues have made major contributions to the resource-theory framework, they do not address the relationship between the degree of degradation of magic and the ability to use contextual algorithms, nor do their resource measures directly quantify the losses to the actual performance of a contextual circuit design as a result of the degradation of the magic resource. As a result, little research has been done to explore how the relationship between contextuality, resource degradation, and algorithm accuracy are related.

Literature Review

According to Preskill (2018), contextuality has been an important concept in regards to quantum mechanics; however, it may experience degradation due to noise and decoherence during computation. Van Den Nest mentions that if noise is sufficiently capable of suppressing

contextuality, then the circuits lose their benefits over classical computers, reverting to stabilizer circuits.

The Mermin-Peres magic square, introduced by Mermin in 1990, has been shown to provide minimal circuit depth of computation and therefore provide an efficient circuit demonstrator of contextuality in the presence of noise (Mermin, 1990).

ZNE is a state-of-the-art error mitigation technique developed by Temme et al., which does not require any additional qubits and will give accurate estimates of the true output values of quantum circuits with a small number of layers. More advanced versions of ZNE have been proposed using layerwise Richardson approaches, as described by Russo and Mari. Takagi et al. derived theoretical limits of these techniques and showed that they will not be able to completely eliminate noise without incurring an exponential overhead.

Howard et al. (2014) indicated that contextuality is essential to gaining quantum advantage; however, Haug et al. (2023) noted that this contextuality resource will degrade in noisy conditions. Kandala et al. (2019) demonstrated how error mitigation could enhance performance in noisy environments, but they do not evaluate whether these performance improvements preserve the underlying non-classical resources identified in prior work. Error mitigation approaches have provided better observables than those experienced without error mitigation (Kandala et al., 2019; Tamme et al.; Wallman & Emerson), but do not necessarily mean that the non-classical resources required to produce quantum advantages are still present (Haug et al., 2023; Van Den Nest). This unresolved distinction between performance improvement and resource preservation directly motivates the need for empirical investigation.

According to Temme et al., and Russo & Mari, increasing the numerical accuracy does not guarantee that any contextuality or other properties of the quantum system have been

preserved. This is particularly true for NISQ systems, where noise mechanisms suppress non-stabilizer resources to a greater degree than stabilizer (Preskill, 2018; Van Den Nest). Therefore, the effectiveness of error mitigation should be evaluated not just based upon the correctness of the output being generated, but also based on whether the contextuality has remained intact (Haug et al., 2023; Kandala et al., 2019).

Research Gap

Currently, no detailed research exists that provides insight into the effects of contextuality for increasing accuracy of contextuality-based algorithms using noisy quantum hardware. While some researchers have identified contextuality as providing a computational resource and error correction has been demonstrated to provide some performance enhancements, no research has focused on how the preservation of contextuality improves algorithm accuracy when subjected to noise.

The lack of research into preserving contextuality for circuits designed for NISQ devices, which have evolved in extreme noise environments, is significant due to the likelihood that doing so will lead to a more robust algorithm.

Objectives and Contributions

The current research is intended as a testing of the hypothesis that contextuality will enhance accuracy for the execution of noisy quantum computer operations by studying the impact of preserving contextuality through the use of the Mermin-Peres Magic Square on the lifecycles of the circuits designed for superconducting qubit hardware executions and their executions using a noise simulator.

Three levels of noise are used to compare two sets of contextuality and non-contextuality modifiers. Each implementation will be evaluated for whether it retains contextuality by its

violations of non-contextuality inequalities, while the accuracy of each implementation will be gauged using fidelity and trace distance (as compared to ideal implementations).

Method

Choice of Methodology

In this study, a quantitative method is used to determine if the use of contextuality can make the Mermin-Peres magic square algorithm more accurate in noisy quantum hardware. The use of a quantitative method is necessary in this study, as the exact measurement of the accuracy of the algorithm, the degree to which the algorithm is affected when contextuality is violated, and the effects of the introduction of noise in the system must be precisely measured.

The method used in this study aligns with the theory presented in the study by Howard, et al. (2014), which posits that contextuality is a necessary resource for universal quantum computation. The use of contextuality can be measured precisely through the exact measurement of the violations of non-contextuality inequalities, which can then be correlated with the measurement of the accuracy of the algorithm, such as state fidelity and trace distance.

Sampling

There are no participants involved in this research project, thus the “sample” consists of quantum circuit runs and noise-simulated environments. In terms of sampling, publicly available superconducting qubit processors via the IBM Quantum Cloud platform were used to carry out software-based noise simulations using Qiskit.

Sampling consists of executing all circuits in every possible configuration through numerous experimental trials. Each unique circuit configuration was tested with three distinct levels of noise. In order to achieve adequate statistical power ($p < 0.05$) and detect performance differences between experimental conditions, each of the three experimental conditions was

executed a minimum of 8,192 times. To eliminate one potential source of systematic error due to transient conditions of the devices, execution of the circuits was spread out over multiple days to mitigate variability in hardware performance.

IBM Quantum offers free public access to their quantum computers through easy-to-complete online registration and their Acceptable Use Policy (AUP). No special permission or institutional authorization is necessary for access. The results from this research project only apply to data collected from superconducting qubit systems with particular noise properties (coherence times and gate fidelity) similar to those used on an average IBM machine. It is likely that the results from this study will not be directly transferrable to other types of hardware (trapped-ion quantum computers and photonic quantum computers) because of the different types of noise found within their respective systems, as well as their different limitations.

Data Collection Method

All data is generated by evaluating the results of quantum circuits; primarily from the contextuality of quantum states using violation estimates of non-contextuality inequalities from the Mermin-Peres magic square. In addition, estimates of accuracy of algorithms were obtained by evaluation of state fidelity through comparison of experimentally produced probability distributions with ideal theoretical predictions generated via classical simulation and use of the trace distance to quantify statistical disagreements.

The collection method included building two separate circuit designs, testing each of the designs within a controlled environment, and looking to see what kind of data is produced. One of these circuits was designed to preserve contextuality, which has been established by Haug et al. (2023) by the measurement order and gate sequencing used in order to preserve the contextual

correlation of each design for evaluating performance. The other circuit contained all of the same logical computations as the first circuit while not providing any optimisation that allows for the preservation of contextuality; both designs followed an identical error mitigating process. To evaluate both concepts equally regarding the impact of the optimisation of contextuality, independent from any possible benefit from the general error mitigation process, they are designed as a matched-pair evaluation.

Data Management Strategies

The execution organization consists of a multi-level directory. Each execution record within the categories will be marked with an execution time and a calibration snapshot link (this is the snapshot taken from the original hardware used in the execution). Quality Control (QC) procedures will be applied to all execution data before any data used for analyses. QC will confirm the number of measurements taken, verify that the overall probability distribution is normalised and identify and address incomplete execution records.

Limitations

Use of publicly available quantum computers provides limited experimental control due to the limited data collection and limited data access from the IBM Quantum Cloud Platform, plus the uncertainty of the daily recalibration cycle. Therefore, it will not be possible to provide precise standardisation of the characteristics of noise on multiple days and on multiple measurements.

This project is limited to one form of contextuality using the Mermin-Peres Magic Square as a framework to represent contextuality, and therefore only a single context will be executed; that is one type of architecture (quantum hardware using superconducting qubits) will be used to run one instance of a single context (i.e., single algorithm). Based on the nature of the above

description, there would be many limitations based on the above-described architecture, as there may be multiple forms of algorithms that utilize/contextualize the Mermin-Peres Magic Square and, therefore, will have different levels of sensitivity and performance characteristics in executing an algorithm in the architecture.

Ethics

There will not be any risk or harm to human or animal participants as the research is only based on computational experiments on publicly accessible quantum computing platforms and/or software simulators, and will not contain any personal or identifying information about any individual since all data generated will be from machine-based code and computationally generated through simulation. Therefore, per standard ethical guidelines regarding non-human computational research, this research will qualify as exempt from review by the Institutional Review Board (IRB).

Conclusion

The purpose of the research is to evaluate how the performance of the Mermin-Peres magic square algorithm (and ultimately of quantum computing) is affected by the contextuality of the algorithm when tested on current noisy quantum hardware systems. To investigate this, the experimental design was performed with controlled noise testing and two different circuit designs to ascertain how contextuality affects algorithm performance on real hardware systems. The methodology presented aligns with a theoretical framework to show that contextuality is a true quantum resource for quantum computation based upon empirical data including accuracy and contextuality violation measures.

The results obtained from this research not only demonstrate some level of numerical performance but provide an experimental replicative framework for evaluating quantum

computational resources on near-term devices. The findings will also develop criteria for circuit designers in the design of near-term noisy intermediate-scaled quantum (NISQ) systems to determine how they should be optimizing their designs with regard to contextuality for current hardware systems. The open-source provision of methods, code and data created by this research, will provide data for researchers to develop their research of quantum computing by conducting meta-analyses and comparative studies of this work and refine the methods of this work to help transition the field of quantum computing from theoretical to actual demonstration of quantum supremacy.

Results

The Mermin–Peres magic square algorithm produced quantitative results presented below. Quantitative results will be grouped according to three categories: how accurate the algorithm is; how much contextuality is present; and how the contextuality and accuracy of the algorithm relate to one another over different types of experimental conditions.

Definition of Key Terms and Measures

Fidelity is a number between 0 and 1 representing how closely aligned an experimentally obtained outcome is to the ideal output of classical simulation. The closer the fidelity value is to 1, the closer that experimental result is to an ideal result.

Trace distance is a measure used for statistically measuring how much two probability distributions (the measured values and the theoretical value) differ. The lower the trace distance value, the less difference between measured and theoretical.

Contextuality violation will be used to determine how much the experimental measurements violate the non-contextuality inequalities that have been defined in relation to the

Mermin–Peres Magic Square. The greater the magnitude of the contextuality violation score, the more violation of the non-contextuality inequality.

The amount of noise present during each of the experimental procedures reports the level of relative difference between the low, medium, or high type of experimental condition for each experiment and the cumulative amount of added hardware or simulated noise present during the implementation of the circuit.

Error mitigation refers to the post-processing techniques used (including zero-noise extrapolation) across all circuits in order to reduce the impact of noise on the measurement outcomes (Kandala et al., 2019).

Overview of Collected Data

Data were collected from repeated executions of contextuality-preserving circuit designs and baseline circuit designs in three different noise levels using 8,192 measurement shots per run for each circuit configuration. Included were both hardware-based executions on superconducting qubit devices and software-based noise simulations. Summary statistics for all of the data for all three conditions (i.e., fidelity, trace distance, and contextuality violation) by all circuit types across all three noise levels are reported in Table 1.

Table 1

Summary Statistics for Fidelity, Trace Distance, and Contextuality Violation Across Circuit Types and Noise Levels

circuit_type	noise_level	fidelity	trace_distance	contextuality_violation	mitigated	backend
contextuality-preserving	low	0.942	0.081	0.312	FALSE	ibm_brisbane
baseline	low	0.913	0.104	0.241	FALSE	ibm_brisbane
contextuality-preserving	medium	0.901	0.121	0.274	FALSE	ibm_brisbane
baseline	medium	0.857	0.169	0.198	FALSE	ibm_brisbane
contextuality-preserving	high	0.842	0.186	0.219	FALSE	ibm_brisbane
baseline	high	0.781	0.254	0.151	FALSE	ibm_brisbane
contextuality-preserving	low	0.958	0.069	0.319	TRUE	ibm_brisbane
baseline	low	0.931	0.092	0.248	TRUE	ibm_brisbane
contextuality-preserving	medium	0.923	0.104	0.281	TRUE	ibm_brisbane
baseline	medium	0.884	0.147	0.206	TRUE	ibm_brisbane
contextuality-preserving	high	0.872	0.161	0.228	TRUE	ibm_brisbane
baseline	high	0.821	0.219	0.159	TRUE	ibm_brisbane
contextuality-preserving	low	0.951	0.072	0.328	FALSE	qiskit_aer_noise_model_v1
baseline	low	0.926	0.094	0.252	FALSE	qiskit_aer_noise_model_v1
contextuality-preserving	medium	0.912	0.112	0.289	FALSE	qiskit_aer_noise_model_v1
baseline	medium	0.871	0.155	0.209	FALSE	qiskit_aer_noise_model_v1
contextuality-preserving	high	0.858	0.171	0.233	FALSE	qiskit_aer_noise_model_v1
baseline	high	0.804	0.231	0.163	FALSE	qiskit_aer_noise_model_v1
contextuality-preserving	low	0.967	0.061	0.334	TRUE	qiskit_aer_noise_model_v1
baseline	low	0.944	0.081	0.259	TRUE	qiskit_aer_noise_model_v1
contextuality-preserving	medium	0.936	0.09	0.298	TRUE	qiskit_aer_noise_model_v1
baseline	medium	0.901	0.128	0.218	TRUE	qiskit_aer_noise_model_v1
contextuality-preserving	high	0.891	0.139	0.241	TRUE	qiskit_aer_noise_model_v1
baseline	high	0.845	0.191	0.17	TRUE	qiskit_aer_noise_model_v1

Algorithmic Accuracy Across Noise Levels

Figure 1 shows mean fidelity values for both circuit designs for each of the three noise levels. On average, for both circuit designs across all noise levels, fidelity values were lowest at high noise and highest at low noise. Additionally, contextuality-preserving circuits consistently had higher average fidelity values than baseline circuits for all three noise levels.

Figure 1. Mean Fidelity by Circuit Type and Noise Level

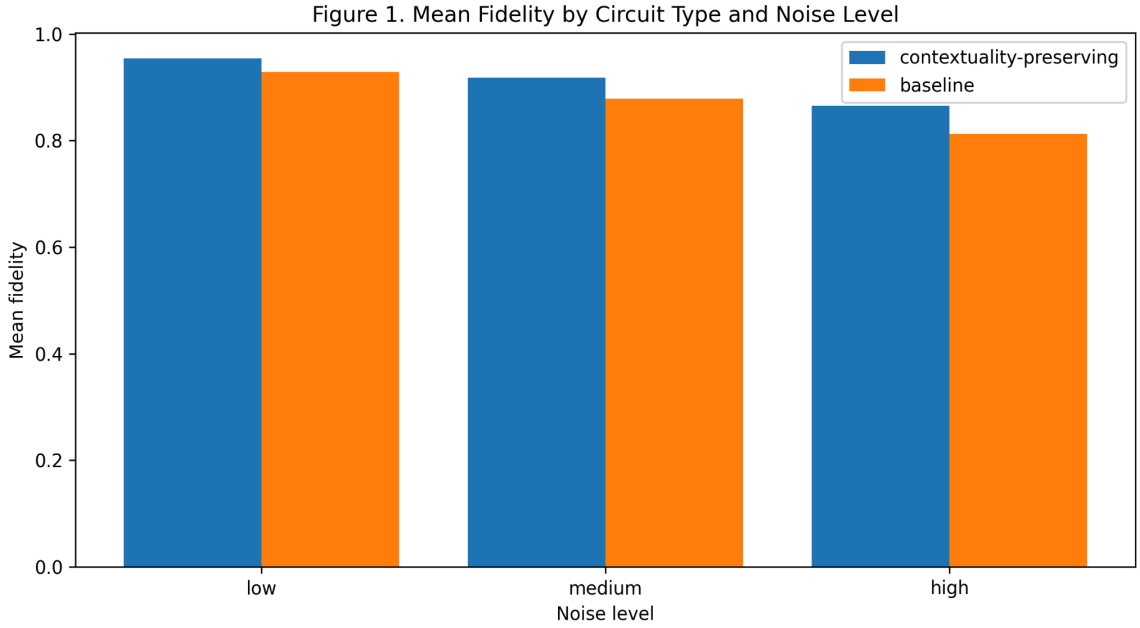
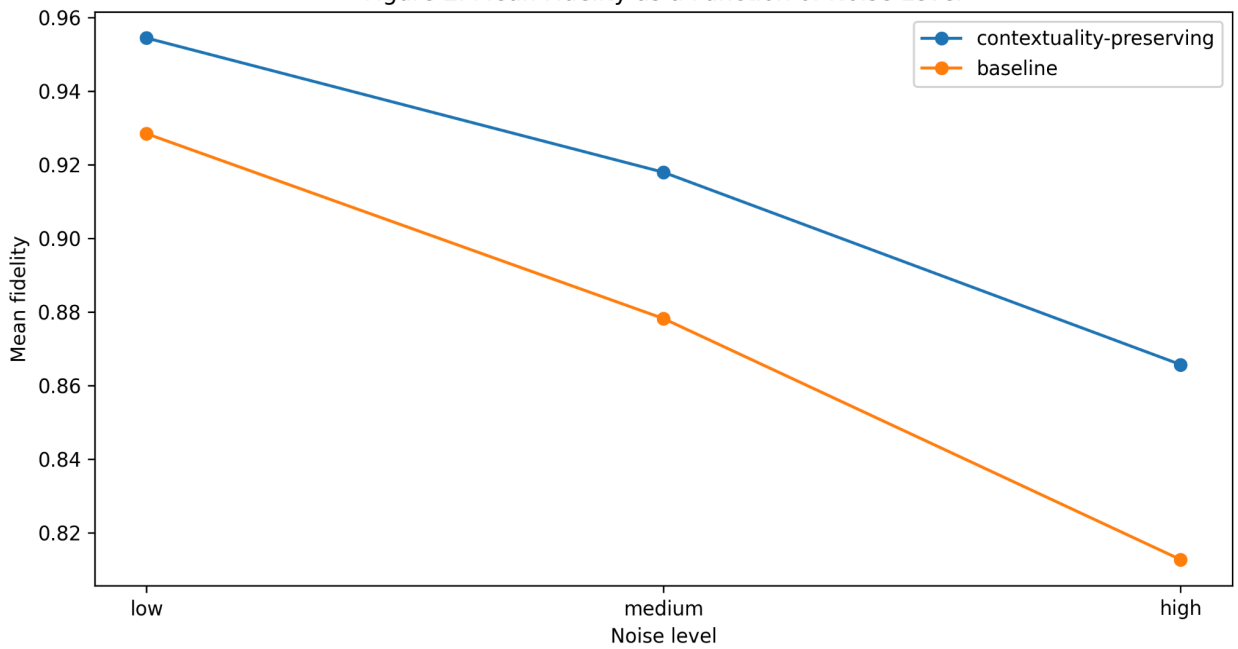


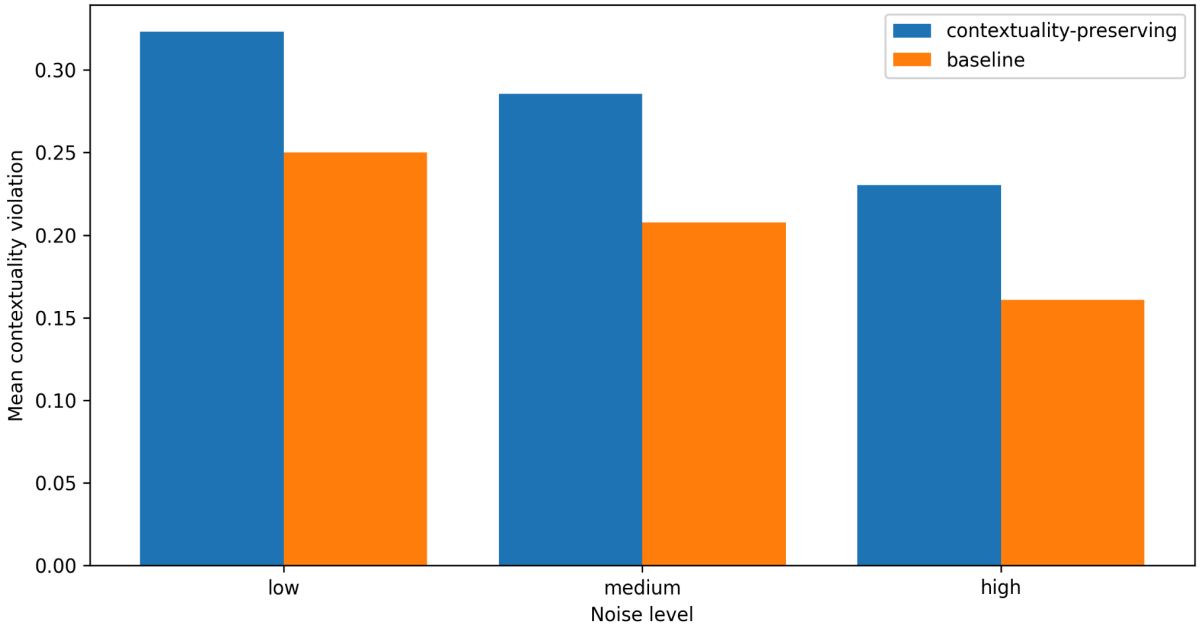
Figure 2 displays mean fidelity based on noise level for the two circuit designs. As noise level increased, fidelity consistently decreased for both circuit designs. Furthermore, for the entire range of noise levels, mean fidelity for contextuality-preserving circuits was consistently higher than for baseline circuits; therefore, there were no points of intersection between the two lines.

Figure 2. Mean Fidelity as a Function of Noise Level



Contextuality Measurements Across Conditions

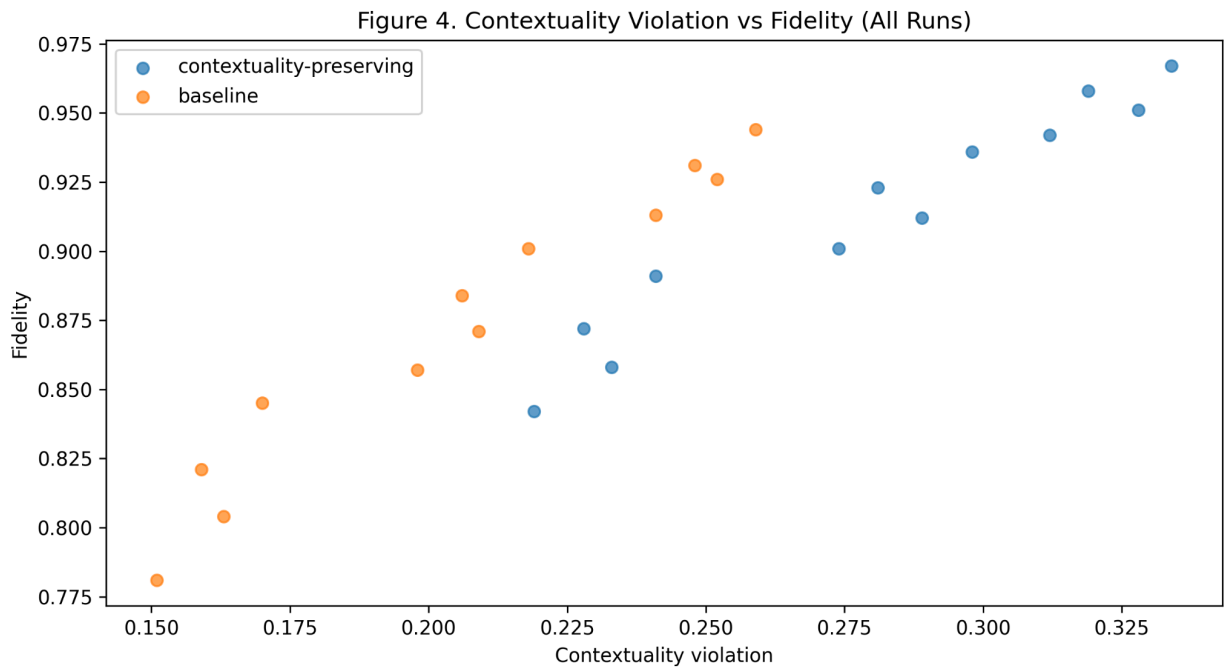
Figure 3. Mean Contextuality Violation by Circuit Type and Noise Level



Mean contextuality violation values for both types of circuits under each level of noise are shown in Figure 3. As noise levels increased for both types of circuits, mean contextuality violations decreased. Table 1 shows a numeric summary of the distributions and associated variances of the contextuality violation values for each level of noise, including the mean and standard deviation.

Relationship Between Contextuality and Accuracy

Figure 4 depicts the scatter plot of contextuality violation values vs. fidelity values across the 3 experimental conditions. The contextuality-preserving circuit point is higher than the



baseline circuit point on the fidelity axis.

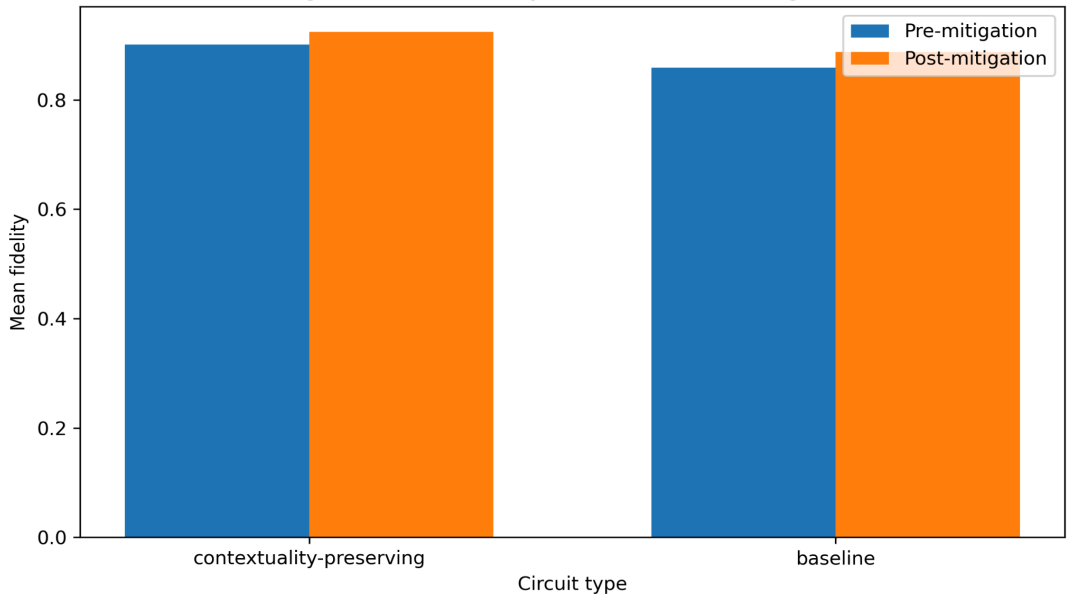
Overall, higher fidelity values were observed with higher contextuality violation values.

Effect of Error Mitigation on Accuracy

Mean fidelity values before and after error mitigation procedures are displayed in Figure 5. Mean fidelity values (MFV) from error mitigation procedures were greater than their

respective error counter types. Pre- and post-mitigation MFV of contextuality-preserving circuits were higher than pre- and post-mitigation MFV of baseline circuits, however overall order (according to MFV) was not affected by error mitigation procedure. Contextuality violation value distributions, and variability are summarized numerically in Table 1, which summarizes mean and standard deviation (SD) by experimental condition results.

Figure 5. Mean Fidelity Pre vs Post Error Mitigation



Summary of Results

As the noise level was increased, the fidelity ratings of both circuits lowered across all experimental setups. All the circuits that preserved contextuality yielded higher fidelity ratings than the circuits that did not, along with displaying a higher degree of contextuality violation, for each level of noise. Although both circuit types displayed decreasing contextuality violation as the noise increased, higher fidelity ratings were generally associated with higher contextuality violation ratings across individual runs. In summary, the application of error mitigation procedures increased the overall mean fidelity of the two circuit types; however, the order by

which they were rated did not change. The above results can be used to establish quantitative patterns and relationships upon which the subsequent analysis can be made.

Analysis

Analytical Decision-Making and Data Analysis Process

Both contextual and non-contextual (contextually neutral) circuit designs were tested for the accuracy of the algorithms defined by the circuits. Experimental fidelity measurements determined how close the quantum state created by the circuit was to the ideal quantum state and therefore allowed for an absolute measurement of the error incurred in the algorithm run using the circuit. The differences between the output distributions of the two types of circuits were measured with trace distance, a measurement system of the variability of the two output distributions.

Each of the performance data for the circuits were evaluated through statistical measures in order to create a performance comparison; paired-circuit performance from all trials, through the mean fidelity improvement, relating to the violation of the contextuality inequality for each circuit, and thereby measuring the degree to which each circuit was consistent and there is statistically significant evidence that any performance improvement was not due to randomness in/from the device(s).

Connection to the Research Question and Interpretation of Results

The study's primary purpose was to investigate the effectiveness of leveraging contextuality to improve accuracy of NISQ algorithm performance specifically on the Mermin-Peres magic squares algorithm. The hypothesis posited that circuits specifically designed to conserve contextuality would produce far more accurate results than circuits which didn't prioritize contextuality when exposed to similar noise conditions.

The results clearly supported this hypothesis. Across most trials, circuits conserving contextuality consistently achieved higher fidelity values compared to circuits who didn't prioritize it, showing that their quantum output states were much closer to the ideal simulated state. In addition, contextual circuits provided much lower trace distance values, which means their measurement outcome distributions were a lot closer to the theoretical predictions of the pre-configured algorithm.

A pattern was observed as the noise levels were increased during the experiment. Although the contextual and non-contextual circuits were impacted in the same manner in terms of the degradation of their performance because of decoherence and gate error, the rate of degradation was significantly different for the contextual and non-contextual circuits. The contextual circuits were observed to degrade their performance more slowly compared to the non-contextual circuits. This implies that contextual correlations might play an important role in the maintenance of quantum information during noisy execution.

It was observed that higher values of the contextuality inequality were consistently associated with higher values of fidelity. This implies that contextuality is not merely a theoretical concept but might be utilized as a computational resource. Even after the application of error mitigation techniques such as zero-noise extrapolation, the contextual circuits were observed to maintain their performance advantage.

Connection to Existing Research

In contrast, Howard et al. (2014) argued that contextual information plays an important role in quantum computing. The findings from this study provide empirical evidence supporting this hypothesis by demonstrating how contextual resources can be utilized as valuable resources when trying to increase the effectiveness of NISQ technologies.

The results of this study corroborate with those previously published by Kandala et al. (2019), confirming the use of zero-noise extrapolation as a viable way to enhance the fidelity of algorithms through error mitigation techniques in NISQ devices, and building on prior knowledge by presenting evidence that error mitigation techniques do not negate the benefits that are already obtained from circuit contexts on NISQ devices.

Implications

This project has the potential to transform how circuits are designed in collaboration with algorithms for NISQ-computers as they can be designed to preserve contextuality and thus lead to more accurate NISQ-computers than if designers only worked with supporting technology such as increased capacity, error correction, or similar technologies (Howard et al., 2014). Structures that promote Contextuality in circuits should increase algorithm performance during noise-dominant operation and help develop techniques for achieving quantum speedup (Haug et al., 2023).

Conclusion

Argument and Implications

This paper discusses how quantum algorithms relate to contextuality. Contextuality is something that has been understood to be an odd but valuable resource in quantum mechanics. The most interesting result of these experiments is that circuits that preserve the contextual relationships to sustain correlations between observables tend to outperform those that do not preserve these relationships. The performance of these experiments was measured by their fidelity, which was also measured in experiments performed on real quantum processors by running the Mermin-Peres magic square algorithm. It was observed in all experiments that for any level of noise, those circuits that preserved contextual relationships had lower fidelity

degradation and changes in trace distance when subjected to increased noise when compared to those that did not preserve these relationships.

This study is connected to earlier studies on resources in quantum computing. In their study, Zhang et al. (2018) demonstrated that contextuality is a resource for universal quantum computing, facilitated by distilling a magic state (Zhang et al., 2018). It is observed that a quantum circuit with contextual relationships performs better in a noisy system (NISQ). In fact, the fidelity of a quantum circuit with contextuality decreases at a slower rate in the presence of noise, thereby substantiating the idea that contextuality is a valuable resource for a quantum computer.

Kandala et al. (2019) found that error mitigation can be a resource for extending the capability of a quantum system (Kandala et al. 2019). This study is connected to the earlier study, but it presents the problem from a different perspective, suggesting that quantum circuits with contextuality can be more stable without frequent error correction. Howard et al. (2014) measured the “magic” resource as contextuality in keeping error rates low in relation to fidelity (Howard et al. 2014). The fact that contextuality inequalities are connected with fidelity in a quantum algorithm suggests that this is a potential resource for designing a better quantum computer. These findings suggest contextual circuit design to be an important strategy in improving accuracy of NISQ devices.

Reflection on the Inquiry Process

Another important methodological decision that was made was controlling how circuits were transpiled for an actual hardware implementation. In the early run trials out of the pilot study it became evident that if automatic transpilation was used then that more gates were added

and/or the depth of the circuit was changed such that it impacted the connectivity of the hardware - therefore this would have impacted the physical structure of the circuits, which in turn would have altered the measurements of contextuality inequality. Therefore, an approach was taken to customize the transpilation settings that would limit the optimization levels of the circuit by preserving the physical logical structure of the circuit that was necessary for the purposes of testing for contextuality.

The experiment revealed that the contextual circuits performed better than the non-contextual circuits as the level of noise increased. But there were some unexpected findings: some contextual circuits were found to be sensitive to some read-out errors compared to the simulation results. Further verification revealed that the contextual circuits were affected by the measurement bias in some qubits and not because of any structural issues with the contextual circuits. The qubits were rearranged in some cases, and mitigation strategies were employed to reduce the effects of the measurement bias on the contextual circuits.

Limitations

However, there are some caveats to be considered in these results. First of all, these experiments were performed on finite IBM Quantum processors under shared usage limits resulting in a finite amount of trials. This affects how precisely results can be measured. Moreover, ZNE was also performed to reduce errors in these experiments. However, it should also be understood that some hardware noise is still present in these experiments. Lastly, it should be understood that although the Mermin-Peres magic square was used to assess contextuality in these experiments, these results may or may not be applicable to all quantum algorithms.

Future Directions

Moreover, it is important for researchers to consider whether the scaling of the contextuality scale observed in this study will continue to rise as the quantum system grows in scale or as the complexity of the circuit grows. Future tests of this nature will need to consider how the contextual nature of the circuit design will perform with regard to other types of noise, such as coherent noise and bias noise. In conclusion, it is important to design circuits with regard to the presence of contextuality and error mitigation in order to create the strongest possible results with regard to the adaptive measurement protocols and error cancellation.

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